

GEOGRAPHY

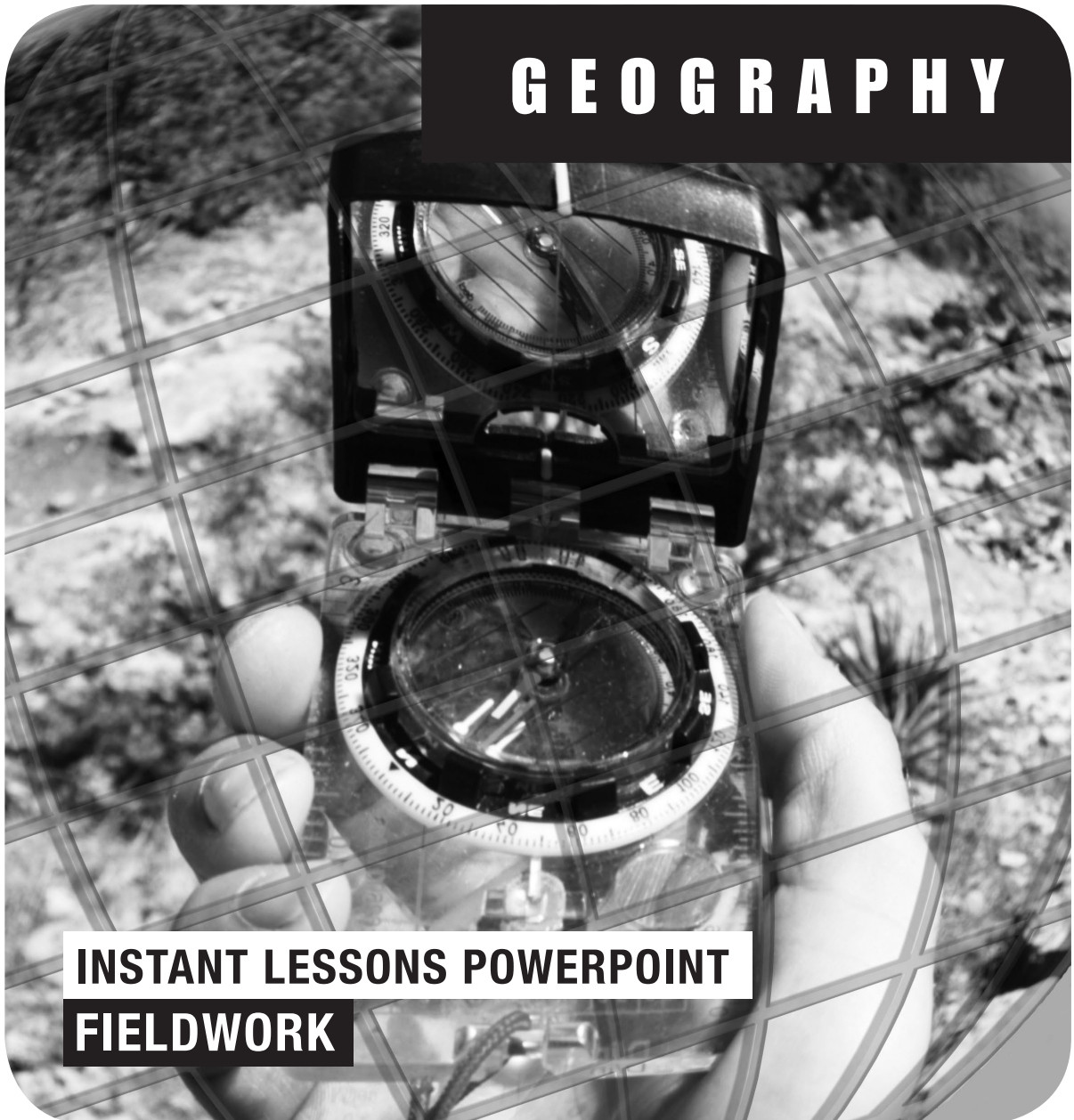


INSTANT LESSONS POWERPOINT

FIELDWORK

Ally Chumley





GEOGRAPHY

INSTANT LESSONS POWERPOINT FIELDWORK

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project management was by Piper Print Media and
Gail McManus Graphics designed the cover.

Ally Chumley is a secondary teacher specialising in English, History and Geography. Her formal credentials include the degrees of Bachelor and Master of Education. Ally has written educational resources for teachers and learners across a range of subjects. She enjoys creating books that make difficult subjects accessible and inspiring to young people.

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EDUCATION
Better ways to learn

Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
info@blake.com.au
www.blake.com.au



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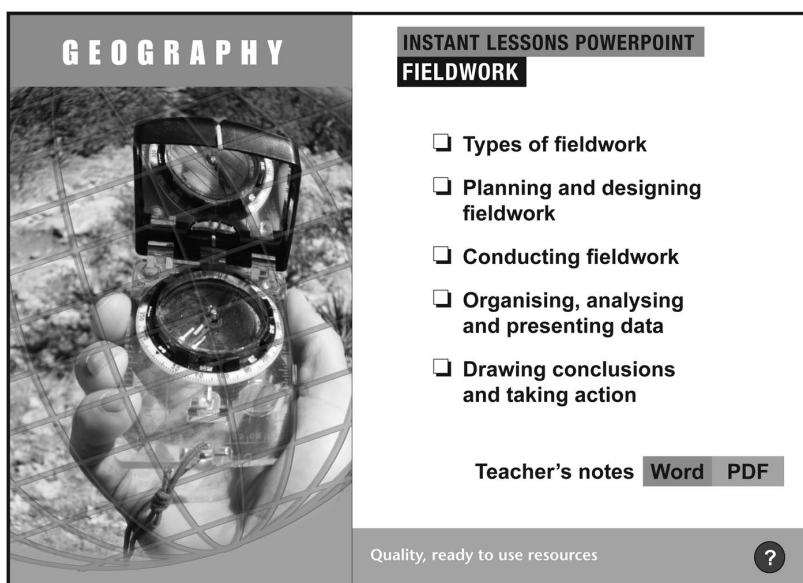
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INTRODUCTION

Instant Lessons PowerPoint slide presentations have been designed to assist teachers in presenting essential geography curriculum to junior and middle secondary school students. Physical, environmental, social and political geography themes are included throughout the series.

What's on the CD-ROM in this book?

- 5 PowerPoint slide presentations
- the teacher's notes and activity sheets in this book, as an editable Word document
- the teacher's notes and activity sheets in this book, as a PDF document.



Key features and uses:

- ✓ Each topic is explored through a core set of slides.
- ✓ Teachers can tailor-make their own slide presentations by adding, deleting or altering slides.
- ✓ Carefully chosen diagrams, maps, graphs, illustrations and photographs illustrate the concepts being taught.
- ✓ On-screen activities test and extend student comprehension.
- ✓ Teacher's notes are provided in this book for each topic.
- ✓ Student activity sheets have been designed so that they can be used during the presentation or later in the classroom, or as homework, extension, assessment or revision. They can be accessed on the CD-ROM in PDF and Word formats, or photocopied from this book.

Contents

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It is recommended that teachers preview the slide sets before showing them to their class, to become familiar with the format and features.

Read the Teacher's notes for an overview of each topic, including key terms and concepts.

The PowerPoint presentations

Starting a slide show presentation

The slide show is used to display the presentation on the screen. It is recommended that you use an external projector or IWB for whole-class use.

There are a number of ways to begin a slide show presentation. In the Windows environment, the CD-ROM will autorun. From the Main menu, simply click on the presentation you wish to view.

In the Macintosh environment, double-click the 'Geography-FW' icon. From the Main menu, click on the presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

Navigating

Main menu

- ◆ From the Main menu, click Close to quit.
- ◆ During the presentations, click Close in the top right corner at any time to return to the Main menu.

Navigating using the mouse

- ◆ Click – Moves presentation to the next slide.

Moving forward

- ◆ Press the N key.
- ◆ Press the Spacebar or Enter key.
- ◆ Press the Page Down key.
- ◆ Press the Right Arrow key.

Moving backward

- ◆ Press the P key.
- ◆ Press the Backspace key.
- ◆ Press the Page Up key.
- ◆ Press the Left Arrow key.

Navigating using the slide show toolbar

At the bottom left corner of the slide show, a toolbar is displayed. There are three buttons that deal with navigation.

- ◆ Left Arrow button – Moves presentation back to the previous slide.
- ◆ Right Arrow button – Moves the presentation to the next slide
- ◆ Slide Icon button – Provides a pop-up menu with Next and Previous menu items, as well as a Go to Slide item that allows for navigation to a specific slide.
- ◆ Right-Click – Provides a pop-up menu with Next and Previous menu items, as well as a Go to Slide item that allows for navigation to a specific slide.

Navigating to a specific slide

There are a number of different ways to display a specific slide.

- ◆ Slide Number + Enter – While in the slide show mode, type in the slide number to be displayed and press Enter.
- ◆ Control-S – To display a menu of all slides in the presentation, press Control and S keys. A menu will appear. Select the desired slide from the menu and press Enter.
- ◆ Slide Icon button – Choose the Slide Icon button from the slide show toolbar. Choose Go to Slide, then choose the desired slide.
- ◆ Right-Click – Choose Go to Slide, then choose the desired slide.

Blanking out the screen

To present a blank screen while in the slide show mode:

- ◆ Slide Icon button – Choose the Slide Icon button from the slide show toolbar. Choose Screen menu item. Select Black Screen or White Screen.
- ◆ Right-Click – Choose Screen menu item. Select Black Screen or White Screen.

Pointer / pen

The Pointer and Pen tools can be used in the slide show view to highlight or draw on the screen, much as would be done on a whiteboard. Drawings can be saved with the presentation or discarded.

Note that while using Pointer tools during a slide show, you will not be able to click the mouse to move to the next slide. Instead, use one of the alternative navigations provided on the previous page, such as the Right Arrow key.

To activate the Pointer tools:

1 In slide show view, right-click, choose Screen | Pointer Options (or click on the Pen icon in the slide show toolbar) and choose one of the following.

- ◆ Arrow – provides an arrow on the screen.
- ◆ Ballpoint Pen – used to write notes on the screen.
- ◆ Felt Tip Pen – a wider pointed pen, useful for drawing. (Windows Office 2003 or higher only)
- ◆ Highlighter – used to apply a highlight of colour that is transparent. (Windows Office 2003 or higher only)
- ◆ Ink Colour – Choose the colour to be used with the tool.
- ◆ Eraser – used to erase entries. (Windows Office 2003 or higher only)
- ◆ Erase All Ink on Slide – removes all entries from the screen. (Windows Office 2003 or higher only)

2 Use the mouse to draw or write on the screen.

Ending a presentation

Each of these actions will end the presentation:

- ◆ Click Close in the top right corner.
- ◆ Press the Esc key.
- ◆ Right-click and choose End Show.
- ◆ Choose the Slide Icon button in the slide show toolbar & select End Show.

Technical specifications

Windows

The CD-ROM will autorun in the Windows environment.

To view the presentations – Windows XP or later operating system

To edit a presentation, open it from within the PowerPoint application.

Note that if editing these PowerPoint presentations in a version prior to Office 2000, cosmetic losses may occur and visual effects such as drop shadows may be lost. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Macintosh

Double-click the file 'Geography-FW' icon on the CD-ROM. From the Main menu, click on the presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

To view a fully featured presentation, and edit it, you must have the PowerPoint 2004 application installed. (i.e. Office 2004 for Macintosh or later)

To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a Macintosh environment, cosmetic losses will occur in some animations, and visual effects such as drop shadows may be lost. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Teacher's notes on the CD-ROM

To open these files you need:

- ◆ **PDF file** – Adobe Acrobat Reader
- ◆ **Word document** – Microsoft Office 2000, or higher.

TOPIC 1: Types of fieldwork

This series of slides introduces the concept of fieldwork and outlines the different types of fieldwork. The tools used in fieldwork and the processes involved are also examined.

Key terms: primary sources, secondary sources, measurement, analysis, statistics, prediction, impact, human environment, physical environment, ethics, investigation, relationships, classify, evaluate, assess

Concepts explored:

- What fieldwork is and what it involves
- Types of measurements taken in meteorology and navigation fieldwork
- What samples are and why they are taken
- Calculations required for creating and interpreting maps
- Using data to predict natural events and assess impacts
- The ethics of gathering people's opinions
- Factors affecting the design of fieldwork
- Classification and finding relationships
- Primary data that can be collected about weather
- Geographical instruments
- Processing and presenting observations

Activity sheets

- Use *Types of fieldwork 1a* and *1b* to test student understanding of the concepts explored in the slide show.
- *Types of fieldwork 2* provides the opportunity for students to apply these concepts to new scenarios.

TOPIC 2: Planning and designing fieldwork

This series of slides works through the steps involved in designing fieldwork for a variety of purposes. It also consolidates ideas about what fieldwork can be used for and why it is important.

Key terms: aim, design, hypothesis, impact, models, simulations, evidence, research, investigation, data types, data collection, risk assessment, organise, analyse, present, findings, report, built environment, comparisons, explain, evaluate

Concepts explored:

- How to determine the best design for fieldwork
- Identifying a clear aim
- Common aims of fieldwork in the physical environment
- Common aims of fieldwork in the human environment
- Different types of resources
- Framing key focus questions
- Methods of collecting data
- Preparing a risk assessment
- Methods of processing data
- Methods of presenting findings and reports
- Studying the built environment
- Why fieldwork is useful

Activity sheets

- Use *Planning and designing fieldwork 1a* and *1b* to test student understanding of the concepts explored in the slide show.
- *Planning and designing fieldwork 2* provides the opportunity for students to apply these concepts to new scenarios.

TOPIC 3: Conducting fieldwork

This series of slides explores in further detail the processes involved in conducting fieldwork, such as collecting primary source material, using secondary sources and using ICT resources to collect data.

Key terms: biosphere, hydrosphere, atmosphere, lithosphere, natural processes, human-activated processes, primary sources, secondary sources, observations, satellite photography, Global Positioning System (GPS), Geographic Information Systems (GIS)

Concepts explored:

- The four spheres of the physical world
- Using fieldwork to explain, predict and evaluate impacts
- How to collect research data
- Making observations – clinometer, compass, theodolite, vernier callipers
- Making observations – digital information
- Methods of collecting primary source material
- Types of secondary source material
- Types of ICT resources
- Using ICT resources to collect data

Activity sheets

- Use *Conducting fieldwork 1a* and *1b* to test student understanding of the concepts explored in the slide show.
- *Conducting fieldwork 2* provides the opportunity for students to apply these concepts to new scenarios.

TOPIC 4: Organising, analysing and presenting data

This series of slides guides students through the process of choosing from the many options available to effectively organise data, analyse it appropriately and present the data in formats that are suitable for both the findings and the intended audience.

Key terms: organise, collate, compile, analyse, connections, patterns, relationships, mathematical tools, statistical tools, interpretation, bias, visual organiser, ICT presentation methods, storage, security

Concepts explored:

- Types of data
- Considerations when organising data
- Methods of organising data
- Using analysis to show patterns and relationships
- Considerations when analysing data
- Methods of analysing data
- Avoiding bias when interpreting results
- Mathematical processing methods
- Statistical comparisons
- Summarising and presenting research findings
- Using visual organisers to present data
- Types of visual organisers and graphs
- Using ICT to present data
- Secure data storage

Activity sheets

- Use *Organising, analysing and presenting data 1a* and *1b* to test student understanding of the concepts explored in the slide show.
- *Organising, analysing and presenting data 2* provides the opportunity for students to apply these concepts to new scenarios.

TOPIC 5: Drawing conclusions and taking action

This series of slides uses the concept of developing a research action plan to explore the issues associated with drawing conclusions, communicating conclusions and using these conclusions to act on the findings and to determine directions for future research.

Key terms: conclusion, ecosystem, accuracy, public awareness, research action plan, ethics

Concepts explored:

- Interpreting findings
- Drawing conclusions
- Considering further implications
- Identifying when further research is needed
- Methods of sharing fieldwork conclusions
- Identifying directions for extending the research
- Developing a research action plan
- Ethics in research

Activity sheets

- Use *Drawing conclusions and taking action 1a* and *1b* to test student understanding of the concepts explored in the slide show.
- *Drawing conclusions and taking action 2* provides the opportunity for students to apply these concepts to new scenarios.

TYPES OF FIELDWORK 1A

An introduction to fieldwork

Fieldwork refers to the activities and tools geographers use to collect data about the world, its life forms and systems. It is completed 'in the field'.

- 1 What does 'in the field' mean in this context?

- 2 Fieldwork usually occurs in a laboratory. True or false? _____

Types of fieldwork

One group of fieldwork activities is the taking of measurements.

Measurement

- 3 Apart from taking readings from measuring equipment, what are two other geographical activities that require the use of measurements? Complete the letter blanks.

a collecting s _ _ _ _ _ s

b creating and interpreting _ _ _ s

- 4 Complete these groups of words about fieldwork techniques in mapping.

a d _ s t _ n c _

k _ l _ _ _ t r _ _

p _ _ x _ m i _ y

b e l _ v _ t i _ n

a _ g _ e

a _ _ i _ _ d e

c l _ n g _ t _ d e

b _ _ _ i n g

d i _ _ c t _ _ _

d s c a _ _

g _ a _ i _ n t

i _ c l _ n e

- 5 Unscramble the letters to find the answer to each question.

- a** Distances, bearings, wind speed and wind direction are all useful measurements for which activity?

i g a v n a t i o n _____

- b** What do researchers try to identify and measure when collecting samples of natural substances?

g e r i d n e n i t s _____

Working with numerical data

- 6 Use these words to complete the passage below: *statistics, figures, count, number, tools*.

Sometimes geographers need to _ _ _ _ _ things, such as the _ _ _ _ _ of people, animals or plants in an environment. Collecting _ _ _ _ _ and working with _ _ _ _ _ involves the use of a range of mathematical _ _ _ _ _.

Predicting events

Professional geographers need to gather data to help predict the timing of certain natural events.

- 7 Why is it important to be able to predict a natural disaster before it occurs?

TYPES OF FIELDWORK 1B

8 Fieldwork can't help researchers predict earthquakes. True or false? _____

9 List three types of natural disasters that you'd like to be able to predict in order to save many lives.

Assessing impacts

10 What is one way of investigating the extent of a flood in a particular region?

Gathering opinions

11 Gathering people's opinions is a fieldwork activity. True or false? _____

12 What are two ways in which people's opinions can be collected?

13 What name is given to the rules that ensure people who provide data are treated fairly and respectfully?

Describing relationships

14 Geographers sometimes need to describe relationships between things. One relationship that fieldwork could be used to study is between flowering plants and in _ _ _ _ _.

Classifying things

15 Complete the passage by filling in the missing letters.

Geographers look for patterns and groupings in physical and human environments.

Classifying things requires fieldwork activities like _ _ serving, _ _ _ ording, _ _ _ cribing, wr _ ting, su _ _ arising and co _ p _ ring.

Measuring the weather

Fieldwork techniques can be used in meteorology to observe, analyse and forecast the weather.

16 List three instruments that can be used in meteorology to observe, analyse and forecast the weather.

17 Name a weather-related measurement that uses each unit.

a mm _____

b cm _____

c knots _____

d degrees Celsius _____

TYPES OF FIELDWORK 2

Using geographical instruments

You can use a simple device called a ‘rain gauge’ to measure rainfall. The gauge collects rainfall and shows the amount of rain in millimetres that fell in the area during a specified time.

- 1 Explain how a meteorologist could use fieldwork techniques to collect and report information about a town’s total rainfall over a 24 hour period.

- 2 Geographers in Samoa are conducting fieldwork as part of a study of the long term impact of the 2009 tsunami on the regrowth of the important species of coastal mangrove trees. Outline one example of each fieldwork activity listed below that these geographers may carry out. One example has been done for you.

Observing	
Recording	Recording the number of new mangrove plants they find per square metre on the mudflats
Writing	
Summarising	
Comparing	

- 3 List two features of the ocean that a geographer might count when conducting a marine field study.

- 4 Complete the table by marking the tallies to indicate the total number of each bird species found in the research zone. Mark each fifth stroke as a diagonal line through four vertical lines, like this: **||||**

Species	Magpie	Indian myna	Kookaburra	Crow	Pee wee
Tally	III				
Total	3	17	6	9	2

- 5 Why might researchers prefer to collect raw numerical data using a tally like those shown above, rather than more sophisticated methods?

PLANNING AND DESIGNING FIELDWORK 1A

Planning fieldwork

Before conducting fieldwork, it is important to have a specific aim in mind. Some common aims of geographical fieldwork are to answer questions about the physical environment and about the human environment.

- 1** List two topics related to the **physical environment** that fieldwork could be used to investigate.

- 2** List two topics related to the **human environment** that fieldwork could be used to investigate.

Identify resources needed

When you plan fieldwork, list any resources you may need. Study secondary sources to familiarise yourself with the key issues. These may include pictorial sources and written sources.

- 3** List four pictorial sources of geographical data.

- 4** List six kinds of written secondary sources.

- 5** Fill in the missing vowels to complete these types of sources.

a p _ t _ t _ _ ns

b m _ lt _ m _ d _ _ pr _ s _ nt _ t _ ons

c _ n _ cd _ t _ l _ v _ d _ nc _

d c _ _ nc _ l _ pl _ ns

e pr _ v _ _ _ s f _ _ ldw _ rk

f c _ mp _ t _ r _ s _ m _ l _ t _ _ ns

g 3D m _ d _ ls

h s _ rv _ y r _ s _ lts

i n _ ws r _ p _ rts

j l _ tt _ rs

k m _ ps

l _ _ r _ _ l ph _ t _ s

After studying the secondary sources, frame some key focus questions and research information specifically addressing those questions.

- 6** A useful way to begin designing fieldwork is to pose a series of questions about the research topic. List five more words of inquiry that you could use to help you frame research questions. One example has been done for you.

What					
------	--	--	--	--	--

PLANNING AND DESIGNING FIELDWORK 1B

Decide how to collect the data

After you identify the type of information needed, determine how it can be collected.

- 7** What information is included in a risk assessment?
-

Decide how you will process the data

- 8** Complete the following instruction about planning to process data.

Write a d _ _ c _ _ p _ _ _ n or list of the m _ _ _ _ _ s chosen to organise and analyse the i n _ _ o _ _ _ _ _ n collected.

Decide how you will present the findings

Determine how you will present the fieldwork findings and report.

- 9** What is one way in which presenting research findings can be satisfying?
-
-

Studying the built environment

- 10** Geography students also conduct investigations into the built environment. List two features of the built environment at your school.
-

- 11** Studying the impacts of the forces of nature and the activities of humans requires students to make comparisons. What is one simple way to do this?
-

The usefulness of fieldwork

- 12** Complete the following text. Then rewrite it so the bullet points are in order from past to present to future.

Fieldwork techniques can be used to explain and evaluate events that:

- may happen in the _ _ _ _ _.
- _ _ _ _ happened in the _ _ _ _
- _ _ _ currently happening

PLANNING AND DESIGNING FIELDWORK 2

Thinking more about planning fieldwork

- 1** Imagine you are planning to build a town reservoir. What are two basic research questions that will need to be answered?
-
-
- 2** Explain how a researcher could use photography to systematically monitor changes in coastal sand dunes over a period of six months.
-
-
-
- 3** Of the resources listed below, tick those that could help you predict the new shape of a coastline due to rising sea levels.
- a** computer simulations ☐ **b** digital mapping tools ☐
- c** conversations ☐ **d** previous fieldwork findings ☐
- 4** Tick the research tasks that would require the use of field sketches.
- a** finding out how many people use the public bus service in your town ☐
- b** planning to plant twenty Norfolk pine trees in a beachside park area ☐
- c** counting the number of kangaroos killed on a particular highway ☐
- d** marking the borders of an ancient burial site sacred to Indigenous people ☐
- e** noting the changes in the shape of a coastline after a series of storms ☐
- f** finding out how much money the average family spends on electricity bills each year ☐
- 5** Number these investigative questions from 1 (first) to 5 (last) to show the correct order.
- a** What is the main research question to be answered? ☐
- b** What is my hypothesis? ☐
- c** What techniques will I use to collect the data? ☐
- d** What type of data do I need to collect? ☐
- e** What is the issue to be investigated? ☐

CONDUCTING FIELDWORK 1A

Investigating the physical world

Geographers use fieldwork techniques to investigate the four spheres of the physical world.

- 1 Define each of the four spheres of the physical world.
- a Biosphere: _____

b Hydrosphere: _____

c Atmosphere: _____

d Lithosphere: _____

- 2 Complete the table below by listing the features of the spheres in the correct columns.
- insects

• crustal plates

• lake

• nitrogen gas

• minerals

• grass

• fish

• the water cycle

• mantle

• ocean

• bacteria

• human beings

• river

• oxygen gas

• rocks

• carbon dioxide

• helium

• core

• spring water

• vapour

Biosphere	Hydrosphere	Atmosphere	Lithosphere

Making observations in the field

Certain tools enable geographers to make very accurate observations of an area.

- 3 Describe a clinometer and outline what it is used for.
- _____
- 4 Name the three-legged surveying device you can use to calculate the vertical and horizontal angles of topographical features.
- _____
- 5 What does a vernier caliper measure? _____
- 6 Which of the instruments from the previous three questions would you use to:
- a estimate the height of a cliff face? _____

b determine the width of a dinosaur’s jawbone? _____

Collecting primary source material

- 7 Add the missing letters to complete the list of fieldwork activities that involve using primary sources.
- a making and interpreting t _ _ _ _ _ _ _ _ _ _ maps

CONDUCTING FIELDWORK 1B

- b** making field s _ _ _ _ _ at a location
- c** writing f _ _ _ _ notes
- d** collecting s _ _ _ _ _ materials
- e** recording impacts of phenomena on h _ _ _ _ _
- f** administering q _ _ _ _ _
- g** conducting s _ _ _ _ _
- h** conducting i _ _ _ _ _

Using secondary source material

8 Circle the correct term to complete each sentence.

- a** Previous research studies, official reports and environmental impact statements are examples of *secondary sources* / *primary sources* / *global positioning systems*.
- b** News reports, old photographs, books, letters and online material are examples of *primary sources* / *secondary sources* / *field observations*.

Using primary and secondary source material

9 Why is a research journal a good tool for a geographical researcher?

Using ICT resources to collect data

Information and Communications Technology (ICT) offers a range of powerful tools for data collection.

10 Rearrange these words to form the names of five ICT data collection tools: *Geographic maps System satellite Internet Systems photographs digital Global the Positioning Information*.

Write each name beside the correct description below.

- a** A navigational system using Earth-orbiting satellites. _____
- b** Taken from a satellite's position in orbit around the Earth. _____
- c** Created by compiling satellite images into a composite picture. _____
- d** An efficient computerised tool that manages geographic information. _____
- e** Provides access to virtually limitless resources of information. _____

Taking care in fieldwork

11 Answer this question by rearranging the words below:

How can a researcher ensure that the data resulting from a study will be reliable and useful?

always, fieldwork, manner, and, should, in, conducted, a, methodical, be, careful

CONDUCTING FIELDWORK 2

Thinking about conducting fieldwork in the physical environment

1 When investigating the effect that a coal-fired power plant has on a river valley environment, impacts on which of the four spheres would need to be taken into account?

2 The Christmas Bell is an Australian wildflower that is now endangered.

a Explain in point form how you could find out the ideal growing conditions for this plant species by counting, taking measurements and collecting samples.

b Imagine you have conducted some Internet research and located a region where Christmas Bells once thrived but are no longer growing. Describe your plan of action for researching this site and explain how you could add to the data you have already collected.

3 What types of fieldwork could be used to find out how many whales beached themselves on the beaches of Byron Bay NSW in the past decade? Circle all that apply from the list below.

a making field sketches

b reading previous research studies

c taking photographs

d collecting sand samples

e recording incidents

f measuring temperature

g reading news reports

h collecting plant specimens

i studying recorded details

j taking a survey

k calculating the height of a cliff face

l conducting Internet research

4 Which activities would be useful for a researcher working out what types of tree-climbing reptiles are living in a particular national park, and which trees they prefer? Circle all that apply from the list below.

a making field sketches

b reading books

c taking photographs

d collecting lizard droppings

e counting numbers of lizards seen

f catching and tagging lizards

g reading news reports

h collecting plant specimens

i studying recorded details

j interviewing people

k marking locations on a map

l conducting Internet research

Organising data collected from fieldwork

Fieldwork is conducted to collect geographical data. Then the information has to be organised.

- 1** Use these words to complete the sentence below: *analysis, patterns, arranged*.

Data can be _____ so that you can see _____ and other relationships with _____.

Collating and compiling methods

You can collate and compile (group together) data using simple methods.

- 2** What is one way of manually collating survey questionnaire forms?

Analysing data

- 3** To analyse findings, researchers look for certain things in different sections of data. Name one.

- 4** What are two ways in which you can avoid bias in interpreting results?

Analysis methods

Simple mathematics can be used to analyse data. Statistical comparisons can be made by calculating various factors.

- 5** Write each statistical tool beside the correct definition.

Statistical tools: *mean, mode, median, range, standard deviation, proportions*

a Relationships between scores and their increasing or decreasing variations _____

b The middle-range score _____

c The average amount of variation of a score from the mean _____

d The average score _____

e The most common score _____

f The scope (the difference between the lowest and the highest scores) _____

Activities after data analysis

- 6** When investigating how your results compare with those arising from a previous study that is similar, what should a researcher consider that may affect the comparison?

Presenting data

7 Write each method of data presentation under the correct heading below.

- bar graphs
 - mind maps
 - information tables
 - line graphs
 - general drawings
 - maps
- pie charts
 - frequency histograms
 - flow diagrams
 - photographs
 - column graphs
 - composite graphs
- proportional graphs
 - diagrams
 - scattergrams
 - numerical tables
 - information charts
 - field sketches

Pictorial methods	Graphs, charts and tables

Using ICT to present data

8 Unscramble the names of these ICT data presentation methods.

- a a bad seat _____

c MC ROD _____

e hog trap shop _____

g solid shews _____
- b DDV _____

d mud cone st _____

f sheeps tread _____

h o audi life _____

Using ICT to store data

Data can be keyed into a computerised storage system and stored in electronic form.

9 Write the full name of each file format abbreviation listed below. Then list at least one application that opens that file format.

- a doc _____

b pdf _____

c txt _____

d html _____

e jpg _____

Thinking more about ways to present data

1 How might satellite image manipulation be used in presenting findings about meteorology?

2 List five types of Internet media that geographers can use to present research findings.

Circle the correct answer to each of these multiple choice questions.

3 Electronic data can be stored in electronic form using:

- a** a hard drive.
- b** a folder of papers.
- c** a filing cabinet.

4 It is impossible to conduct a fieldwork investigation that produces all the answers to a particular problem because:

- a** it is too expensive to buy the necessary equipment.
- b** there are numerous variables that have differing impacts.
- c** research is not usually carefully planned.

5 What is one way to keep raw data secure so it cannot be tampered with or lost?

6 Draw lines to match each mathematical process with the research topic to which it is best suited.

Mathematical processes

- a** simple counting and comparing of totals
- b** calculating maximum, minimum, range and average
- c** expressing data as percentages
- d** applying mathematical formulae
- e** using statistical comparisons
- f** using existing data to make predictions

Research topics

- *calculating a region's temperature*
- *describing populations of major cities*
- *predicting extinction dates for a species*
- *listing the most popular tourist destinations*
- *estimating the volume of water in an ocean*
- *finding the unemployment figures for a town*

7 Calculating a ratio is a mathematical process that could be useful in data analysis. Yes or no? _____

DRAWING CONCLUSIONS AND TAKING ACTION 1A

Drawing conclusions from fieldwork

When fieldwork has been completed and the data collected, organised and presented, it is time to interpret the findings and draw conclusions about the investigation.

Drawing conclusions

Because geographers study complex environments, reaching accurate conclusions can be difficult if the data are not handled carefully.

- 1 Geographers rarely examine only a single element of an environment. True or false? _____
- 2 What are two questions that geographers can ask to help guide their conclusions?

Further implications

Learning about the physical and human environments is an ongoing process.

- 3 There will always be the need for further geographical research. True or false? _____
- 4 Unscramble the words in bold to complete this sentence. Then rewrite it correctly.
Geographical **seerharc** aims to **velos lemborps**.

Repeating fieldwork

- 5 Fieldwork may need to be repeated for a number of reasons. List two reasons.

When more research is needed

When a study doesn't reach a satisfactory conclusion, further action should be taken.

- 6 What is one question that can help a researcher find a clear direction for further action?

Sharing fieldwork conclusions

- 7 Geographers provide certain groups with information about their research findings. List three of these groups.

DRAWING CONCLUSIONS AND TAKING ACTION 1B

Developing a research action plan

A research action plan is developed in eight stages.

- 8** Break up these word chains into ten separate terms. Then use the terms to complete the steps in creating a Research Action Plan.

ANALYSE CONCLUSIONS ANSWER QUESTION TECHNIQUES
ACTION PURPOSE PRESENTATION COLLECT PRIMARY

- Step 1** Identify the _____ of the research study.
- Step 2** Write focus _____ that need to be addressed.
- Step 3** Identify the _____ and secondary data needed to _____ the focus questions.
- Step 4** Select _____ to use to collect data.
- Step 5** Methodically _____ the primary and secondary data.
- Step 6** Organise and _____ the collected data.
- Step 7** Select methods of _____ for the research findings.
- Step 8** Draw _____, make a plan and take _____ in response to the findings.

Ethics in research

It is essential for researchers to act appropriately when conducting research. These concerns are called 'ethics'.

- 9** Write each example beside the correct field of ethics.
Examples: *accurate reporting*, *preservation of privacy*, *conservation of natural resources*.

- a** Ethics and the physical environment _____
- b** Ethics in research practices _____
- c** Ethics in dealing with people _____

Developing your fieldwork skills

Fieldwork in geography encompasses a whole range of skills, knowledge and activities that all students of geography can master.

- 10** Circle the correct options in italics to complete these sentences.
- a** Skills in planning and conducting fieldwork lead to *positive changes* / *higher profits*.
- b** Conclusions should be *the products of guesswork* / *underpinned by scientific research*.

DRAWING CONCLUSIONS AND TAKING ACTION 2

Developing your fieldwork skills

- 1** Research conclusions can be made available to the general public in various ways. Suggest which method from the list below would be most suitable for presenting findings about each issue.

Methods

- television news reports
- information brochures
- books
- newspapers
- public lectures
- community announcements
- magazines
- official reports

- a** The effects of climate change on your local town _____
- b** The best ways to avoid catching swine flu _____
- c** Statistics on obesity for each Australian state _____
- d** An earthquake warning for the local region _____

- 2** Write each term beside the correct definition. Terms: *tally, statistics, outcome, accuracy, investigate, researched, sample, population*.

- a** correctness _____
- b** small portion of something _____
- c** studied methodically _____
- d** numerical information _____
- e** people living in an area _____
- f** look into _____
- g** count _____
- h** result _____

A research scenario

- 3** Imagine this scenario:

You have just completed a two year research study into the effects of litter in a coastal town. You have determined the following.

- ◆ Tourists are throwing their rubbish around on the beaches, and plastic bags are causing death and injury to various sea birds and other wildlife species. The litter is also causing the area to look unattractive and is clogging up stormwater drains. It is worse in the summer months, which is the peak tourist season.
- ◆ An average of 4 tonnes of rubbish is dumped by tourists throughout the town each summer.
- ◆ Specific ecosystems are being damaged. Osprey nesting grounds are being abandoned. Pelicans are swallowing plastic and need veterinary treatment.
- ◆ Beaches, parks and bush-walking trails near the creek are covered with decaying litter. This causes smells, bacterial growth and water pollution.
- ◆ Local residents overwhelmingly agree that the points above are impacting negatively on the town. They have agreed to take action.

List the fieldwork techniques you could have used to collect information about the following factors.

- a** Amount of rubbish _____
- b** Abandonment of osprey breeding grounds _____
- c** Effect on pelicans _____
- d** Creek water pollution _____
- e** Views of residents _____

SELECTED ANSWERS

Types of fieldwork 1

- 1 In the research situation – the actual environment that the researcher is investigating.
- 2 false
- 3 **a** samples **b** maps
- 4 **a** distance, kilometres, proximity
b elevation, angle, altitude
c longitude, bearing, direction
d scale, gradient, incline
- 5 **a** navigation **b** ingredients
- 6 count, number, figures, statistics, tools
- 7 To warn people to evacuate; to protect or fortify resources, plants and animals, buildings and other built environment features; to prepare emergency services; to find ways of preventing disasters in the future.
- 8 false
- 9 For example: earthquakes, tsunamis, volcanic eruptions, cyclones, tornados, bushfires, landslides, floods, blizzards, extreme weather events
- 10 For example: using aerial photography, surveying people, collecting samples, mapping
- 11 true
- 12 survey questionnaires, interviews
- 13 ethics
- 14 insects
- 15 observing, recording, describing, writing, summarising, comparing.
- 16 For example: rain gauge, thermometer, anemometer, hygrometer, barometer, weather balloon
- 17 **a** rainfall **b** snow **c** wind speed
d air temperature

Types of fieldwork 2

- 1 Using a rain gauge, take measurements every 6 hours to obtain rainfall figures. The data could be presented in various ways, including as a table of measurements, perhaps showing the average rainfall each 6 hours (the total of the figures obtained over 24 hours divided by 4).
- 2 For example:
Observing – noting water levels, tidal markings, damage to the environment, other types of flora present, size of existing trees
Writing – writing down the observations made
Summarising – writing a report that summarises the main findings of the study
Comparing – looking at secondary

sources to compare findings, comparing the number of mangroves growing in a similar area that was unaffected by the tsunami

- 3 For example: numbers of any marine plant or animal species, breeding pairs, distances travelled, rates of decay/degradation, pollution statistics, siltation, salinity and other measures involving minerals

4

Indian myna	Kooka-burra	Crow	Pee wee
HHH HHH HHH II	HHH I	HHH IIII	II
17	6	9	2

- 5 It is a good way of keeping track of which items have been counted. While counting in the field, it's easy to mark a single stroke each time the item is observed, especially if the researcher is moving around a particular location. Tallies can easily be altered by adding more strokes, rather than crossing out and changing fixed numerical totals. The tally marks can be counted up at the end.

Planning and designing fieldwork 1

- 1 For example: ecosystem productivity, extreme weather events, coastal processes and management, climate change
- 2 For example: impacts of industry, city planning, rural-to-urban population shifts, sustainable tourism
- 3 e.g. maps, satellite or aerial photos, ordinary photographs, sketches, diagrams
- 4 e.g. news reports, scientific research reports, letters, survey results, interview transcripts, petitions, council plans and records, books
- 5 **a** petitions **b** multimedia presentations **c** anecdotal evidence **d** council plans
e previous fieldwork **f** computer simulations **g** 3D models
h survey results **i** news reports
j letters **k** maps **l** aerial photos
- 6 where, when, how, who, why
- 7 potential hazards and an outline of a risk management plan
- 8 description, methods, information
- 9 When the results are obtained from data collected by the student themselves, and when the research results will be useful in helping to solve a particular problem in the local area.

- 10 For example: features like buildings, roads, pathways, bicycle racks

- 11 Taking 'before' and 'after' photographs

- 12 Fieldwork techniques can be used to explain and evaluate events that:
 - have happened in the past
 - are currently happening
 - may happen in the future.

Planning and designing fieldwork 2

- 1 For example: Where will the reservoir be built? What water volume will it need to be able to hold? Where is the best location for it? What type, size and extent of piping and pumping will need to be used? What problems need to be overcome to complete the project? What will it cost? How much rainfall does the site receive? What drainage occurs at the site? How will the project impact upon flora and fauna species?
- 2 The researcher could take photographs of the dunes from exactly the same locations and angles each day/week/month for a period of six months. These could then be compared, using tracings and overlays, or digital imaging, to determine the type and rate of change.
- 3 computer simulations and digital mapping tools
- 4 b, d, e
- 5 in order: b = 1, c = 2, e = 3, d = 4, a = 5

Conducting fieldwork 1

- 1 **a** Biosphere: the Earth's living components
b Hydrosphere: the Earth's water resources
c Atmosphere: the layer of gases that are above the Earth
d Lithosphere: the Earth's crust and mineral elements
- 2 Biosphere: insects, bacteria, human beings, grass, fish
Hydrosphere: lake, the water cycle, ocean, river, spring water
Atmosphere: nitrogen gas, oxygen gas, vapour, helium, carbon dioxide
Lithosphere: minerals, core, mantle, rocks, crustal plates
- 3 It is a hand-held tool that helps the user determine whether or not a surface is level.
- 4 theodolite
- 5 small distances between opposing sides of an object
- 6 **a** theodolite **b** vernier calipers

SELECTED ANSWERS (cont'd)

- 7** **a** topographic **b** sketches **c** field
d sample **e** humans
f questionnaires **g** surveys
h interviews
- 8** **a** secondary sources
b secondary sources
- 9** It helps to document their activities and keep track of all source materials used.
- 10** **a** Global Positioning System (GPS)
b satellite photographs **c** digital maps
d Geographic Information Systems (GIS) **e** the Internet
- 11** Fieldwork should always be conducted in a methodical and careful manner.

Conducting fieldwork 2

- 1** All four.
- 2** **a** Locate places where they are currently growing; take soil samples; take samples of surrounding species of flora and fauna; measure temperatures; take water samples by collecting rainwater or nearby water that irrigates the plants; take air samples and test for pollutants.
b Look at secondary sources to research areas in which Christmas Bells used to thrive, but no longer live today. Conduct the same activities at those locations and compare the two sets of data to explain the differences.
- 3** b, e, g, i, l
- 4** a, b, c, d, e, f, h, i, k, l

Organising, analysing and presenting data 1

- 1** arranged, patterns, analysis
- 2** e.g. sorting forms into piles, or using a tally sheet to record responses
- 3** e.g. connections, patterns, other relationships
- 4** A sound research plan, personal awareness and several methods of cross-checking results and findings
- 5** **a** proportions **b** median
c standard deviation **d** mean
e mode **f** range
- 6** The specific conditions and variables in the two studies that may have been different
- 7** Pictorial methods: mind maps, general drawings, flow diagrams, photographs, diagrams, maps, field sketches
Graphs, charts and tables: bar graphs, line graphs, column graphs, composite graphs, proportional graphs, information charts, scattergrams, pie charts, frequency histograms, information tables,

numerical tables

- 8** **a** database **b** DVD **c** CD-ROM
d documents **e** photographs
f spreadsheet **g** slide shows
h audio file
- 9** **a** document; Microsoft Word
b portable document format; Adobe Acrobat Reader **c** text file; Microsoft Word **d** hyper text markup language (Internet coded files); Internet Explorer **e** Joint Photographic Experts Group; Photoshop

Organising, analysing and presenting data 2

- 1** To create composite (layered) weather images of cloud cover, storms or other atmospheric conditions.
- 2** web pages, email, e-publishing (e-books and e-zines), blogging, social networking sites, digital file-sharing sites
- 3** **a**
- 4** **b**
- 5** e.g. keeping electronic data in a secure, password-protected computer file, keeping documents in a locked filing cabinet
- 6** **a** describing populations of major cities **b** calculating a region's temperature **c** finding the unemployment figures for a town
d estimating the volume of water in an ocean **e** listing the most popular tourist destinations
f predicting extinction dates for a species
- 7** Yes

Drawing conclusions and taking action 1

- 1** true
- 2** For example: What do we now know about the issue? Can the research questions now be answered? What are the answers? Which particular findings were expected? Which ones were unexpected? Can any general conclusions be drawn from this study? What are they? Can the research problem now be solved? If not, what should be done about it? What are my feelings about it?
- 3** true
- 4** Geographical research aims to solve problems.
- 5** e.g. in order to show that the original research findings are still accurate; to test the same hypothesis in a new setting; to resolve conflicting results from other

studies; because the research did not result in reliable conclusions.

- 6** e.g. What or who is most affected by this issue? Is fixing this problem likely to bring about a positive change in the life of a person or species?
- 7** e.g. governments, individuals, organisations and the general public
- 8** [in order] purpose, questions, primary, answer, techniques, collect, analyse, presentation, conclusions, action
- 9** **a** conservation of natural resources
b accurate reporting
c preservation of privacy
- 10** **a** positive changes
b underpinned by scientific research

Drawing conclusions and taking action 2

- 1** Answers will vary.
a newspapers, official reports, public lectures, information brochures, community announcements
b information brochures, community announcements, television news reports
c television news reports, newspapers, official reports, magazines, information brochures
d television news reports, community announcements
- 2** **a** accuracy **b** sample
c researched **d** statistics
e population **f** investigate
g tally **h** outcome
- 3** **a** collect and count rubbish from a few sample areas that represent the whole area; use statistics to calculate extent of problem; weigh samples taken and multiply by total surface area affected.
b Use secondary sources to determine usual breeding patterns in previous years; count numbers of nests, breeding pairs and offspring over a two-year period and compare the figures from previous years.
c Interview/survey local veterinarians, wildlife rescue organisations to obtain figures and types of injuries.
d Take water samples from places upstream and downstream of littered areas and analyse the type of pollutants and their concentration, as well as their likely origin.
e Take surveys or conduct interviews with specific questions directed to collecting views.

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Fieldwork

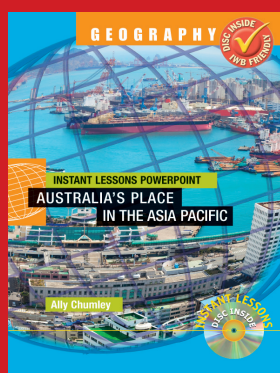
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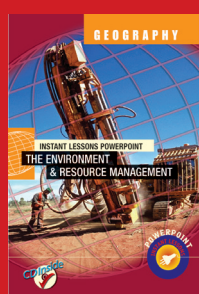
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- 2: Planning and designing fieldwork
- 3: Conducting fieldwork

- 4: Organising, analysing and presenting data
- 5: Drawing conclusions and taking action

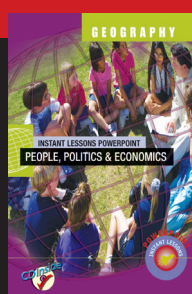
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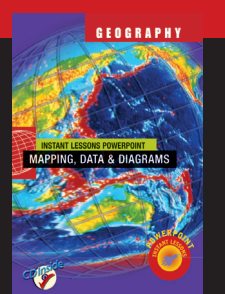
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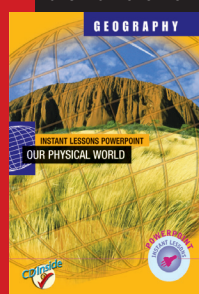
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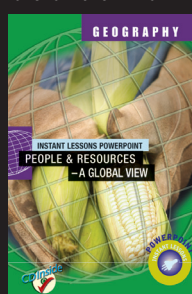
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108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

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